

No One AI Fits All: Technology and Functional Complementarities in Firm-Level AI Adoption (2021–2024)

1.- Introduction

Artificial Intelligence (AI) has rapidly emerged as a transformative force in modern organizations (Agrawal et al., 2019, 2022; Baily et al., 2023; Brock & Von Wangenheim, 2019; Davenport & Ronanki, 2018; Ransbotham et al., 2020;). However, realizing its promised benefits remains a challenge for many firms (McKinsey, 2024, 2025; Ransbotham et al., 2017). While most organizations recognize AI as a strategic opportunity and pursue adoption for competitive advantage, success in capturing tangible business value is far from guaranteed (Chui et al., 2018; de Bellefonds et al., 2024; McKinsey, 2024, 2025; Ransbotham et al., 2020; Tse et al., 2020). According to a recent report by BCG (de Bellefonds et al., 2024), only 26% of surveyed firms have consistently generated substantial value, as measured by revenues, total shareholder return, innovation, and employee satisfaction.

Despite the proliferation of AI investments and high expectations, case studies frequently show that anticipated outcomes—such as process optimization, improved decisions, and strategic transformation—often fail to materialize (Ångström et al., 2023; Ransbotham et al., 2017, 2019, 2020). Persistent obstacles include limited integration of AI into existing workflows, scarcity of specialized skills, insufficient digitization of core processes (Jung and Gómez-Bengoechea, 2025; Schuh et al., 2020; Zolas et al., 2021), and a lack of clarity on how various AI technologies create value (Atkinson & Ezell, 2019). Consequently, many firms experience stagnating or low-impact AI projects (de Bellefonds et al., 2024; Enholm et al., 2022). To unlock AI's potential, firms must identify which technologies best fit their strategic and operational contexts and understand the complementarities among them (Ameye et al., 2023; Davenport and Mittal, 2023; Duan et al., 2019; McKinsey, 2025; Peretz-Andersson et al., 2024).

Although AI's potential to boost innovation, productivity, and competitiveness is widely acknowledged, surprisingly little is known about the factors influencing the use of specific AI technologies and how they are deployed across different functional areas within firms. Most prior studies have treated AI as a single, monolithic technology (Acemoglu et al., 2022; Bresnahan, 2022; Calvino and Fontanelli, 2022, 2023, 2024; Cho et al., 2023; Kinkel et al., 2022; McElheran et al., 2024; Zolas et al., 2020), analyzing its aggregate effect on firm performance. McKinsey (2022) found

that the average number of AI capabilities organizations utilize, including natural language generation and computer vision, has also doubled—from 1.9 in 2018 to 3.8 in 2022. Recently, two studies have filled that gap (Ameye et al., 2023, Calvino & Fontanelli, 2025). Ameye et al., (2023) was the first study to analyze factors affecting the use of some AI technologies, i.e., image recognition, ML, NLG, NLP, robotic process, speech recognition and virtual agents. The second study is the one by Calvino and Fontanelli (2025) who used a large-scale survey for French firms to advance this literature by demonstrating that firms adopt multiple AI technologies for distinct purposes and business functions, emphasizing that “*each AI tool may vary in requirements, serve distinct purposes, and be applied to different business functions*” (p. 3). This finding aligns with McKinsey (2024, p. 3), which asserts “*organizations are using AI in more business functions now than in previous years.*”

Experts agree that AI’s transformative potential extends across all functional areas of the firm, but its implementation is far from a one-AI-fits-all process. Firms must recognize the diversity of AI technologies—such as text mining, speech recognition, natural language generation (NLG), image recognition, machine learning, automation of workflow processes, and robotics—each providing unique advantages and applications across functional areas (Ameye et al., 2023; ; Calvino and Fontanelli, 2025; Chui et al., 2018; Enholt et al., 2022; Gerbert et al., 2017; Haefner et al., 2023; OECD/BCG/INSEAD, 2025). For instance, natural language processing can enhance HR and recruitment (Makarius et al., 2020; Prikshat et al., 2023; Radonjić et al., 2024), employee training (Calvino and Fontanelli, 2024), and customer engagement (Pérez-Vega et al., 2021). These cross-functional applications highlight the economies of scope that emerge when AI technologies are effectively integrated (OECD/BCG/INSEAD, 2025).

However, despite growing interest, little empirical research examines how firms combine different AI technologies or how these technologies interact across internal functions. This gap is striking given the evidence that each AI technology can be used in multiple areas—marketing, HR, logistics, or production—allowing firms to exploit synergies and economies of scope (Ameye et al., 2023; OECD/BCG/INSEAD, 2025). As Chui et al. (2017) observed, “*For AI to realize a substantial portion of the potential value we have estimated will require companies to deploy these techniques comprehensively in areas where they can most effectively harness their ability to make the complexity of data an advantage*” (p. 17). Furthermore, the diversity of AI technologies poses a new level of uncertainty for firms. In the case of a homogeneous ICT use, firms can copy what other firms have done. However, as asserted by Ameye et al., (p. 15) “*when the technology and its potential use cases are so diverse, it is not enough to learn that a competitor has deployed a certain AI with some success,*

one needs to get more details about which variant of the AI has been adopted, in which area of the business, and for what purpose. This makes imitation much less straightforward than with other technologies that have clearer or simpler use cases”.

Given these insights, this study explores two related research questions:

RQ1: Are there interdependencies or complementarities among AI technologies used by firms?

RQ2: Is there interdependence in the use of these AI technologies across functional areas of firms?

Using data from the Regional Center for Studies on the Development of the Information Society (CETIC) in Brazil for 2021, 2023 and 2024, we analyze both the factors associated with the adoption of seven AI technologies—text mining, speech recognition, NLG, image recognition, machine learning, workflow automation, and robotics—and the patterns of AI use across different functional areas, including marketing, production, business administration, management, logistics, digital security, and human resources.

Building on this empirical foundation, our study makes two main contributions to the literature. First, it extends the emerging stream of research on AI adoption by disaggregating artificial intelligence into its constituent technologies—text mining, speech recognition, natural language generation, image recognition, machine learning, workflow automation, and autonomous movement or robotics (Ameye et al., 2023; Calvino and Fontanelli, 2025). By analyzing how firms adopt and combine these technologies, we offer new evidence of technological interdependence and complementarity within the AI ecosystem. This multidimensional perspective helps move beyond the "black-box" treatment of AI as a single, homogeneous technology, instead revealing how firms strategically deploy different AI tools according to their technical affordances, data requirements, and functional purposes.

Second, the paper expands the existing understanding of cross-functional AI integration by examining how firms apply AI technologies across diverse business areas, including marketing, production, logistics, business administration, management, digital security, and human resources. Our findings indicate that larger and more digitally mature firms exhibit both a greater likelihood of adopting multiple AI technologies and a higher degree of functional diffusion, suggesting the existence of economies of scope in AI use. The evidence of interdependence or complementarities between technologies and business functions underscores AI's systemic nature as an organizational capability, rather than a set of isolated tools (Ameye et al., 2023; Calvino and Fontanelli, 2025; McKinsey, 2025).

In doing so, the study contributes to ongoing debates on digital transformation, technological complementarities, and the socio-technical integration of AI in firms, providing an empirical grounding for future research on how AI-driven complementarities shape firm performance and innovation trajectories.

Lastly, the study presents a graphical exercise of the conditional probability of using a given AI technology given the probability of each of the remaining six. Likewise, we examine the conditional probability of using AI in a given functional area, given the conditional probability of using AI in each of the remaining functional areas. We make a comparative dynamic analysis of the co-occurrence of using those AI technologies. The analysis reveals the nuanced relationships between AI technologies and their applications across the functional areas of firms.

The remainder of the paper is structured as follows. Section 2 reviews the existing literature on the adoption of AI, specifically the family of applications and technologies, and their application within various business areas of firms. Section 3 details the data sources utilized in this study—namely, the CETIC ICT survey. In Section 4, we investigate the factors associated with the use of those two digital technologies, focusing on ICT-related complementary factors and conducting the analysis by sector. Finally, Section 5 concludes with a synthesis of the main findings and offers suggestions for future research directions.

2. Literature Review and Theoretical Background

Artificial intelligence (AI) has evolved into a transformative family of technologies that collectively redefine how firms create, coordinate, and capture value (Akkiraju et al., 2020; Collins et al., 2021; Uren & Edwards, 2022). Broadly understood as the development of computer systems capable of performing tasks that typically require human intelligence—such as learning, reasoning, and perception (McCarthy, 2007; Russell and Norvig, 2021)—AI today spans a diverse set of tools and applications including text mining, speech recognition, natural language generation (NLG), image recognition, machine learning (ML), workflow automation, and robotics and autonomous movement. These technologies differ in their technical foundations and areas of application, but they increasingly interact in complementary ways, shaping a multidimensional digital transformation across all functional areas of the firm (Calvino and Fontanelli, 2025; OECD/BCG/INSEAD, 2025).

2.1. A Socio-Technical Systems Perspective on AI Adoption

Socio-Technical Systems Theory (STST) provides a holistic framework for understanding AI adoption as the outcome of interactions between technological and social subsystems within organizations (Haefner et al., 2023; Makarius et al., 2020; Sony & Naik, 2020). From this perspective, AI adoption entails not only technological deployment but also the reconfiguration of human skills (Lee et al., 2022; Calvino and Fontanelli, 2024), organizational routines (Ameye et al., 2023), and managerial practices (Fountaine et al., 2019) to ensure joint optimization between technical and social systems (Ångström et al., 2023; Krakowski et al., 2025).

Applied to AI, STST explains how diverse technologies -ML algorithms, automation systems, text and image recognition tools, or NLG- can only generate value when supported by appropriate social structures such as training programs, leadership support, and adaptive work processes (Anthony et al., 2023; Uren & Edwards, 2023). Moreover, it emphasizes that firms' capacity to scale AI depends on the mutual alignment of digital infrastructures and organizational culture, which together enable learning, experimentation, and integration across departments. In this sense, the interdependencies among AI technologies mirror interdependencies within socio-technical systems: both rely on complementary resources, information flows, and coordination mechanisms to produce collective intelligence.

2.2. Complementarities among AI Technologies and Dynamic Capabilities

Recent studies conceptualize AI not as a single invention but as a system of interdependent technologies whose combined use generates synergistic benefits (Ameye et al., 2023; Brynjolfsson & McElheran, 2019; Collins et al., 2021; Holmström, 2022). Each technology contributes distinct yet mutually reinforcing functionalities—text mining, speech recognition, and NLG enable firms to process and generate language-based data; image recognition supports pattern detection in manufacturing and logistics; ML provides the analytical backbone for prediction and insight generation; and automation and robotics operationalize these insights in workflows and production systems (Ulrich and Frank, 2021; Wong et al., 2025). These interdependencies create strong technological complementarities, where the value of one technology increases with the presence of others (Brynjolfsson and Milgrom, 2013; Ennen and Richter, 2010; Milgrom and Roberts, 1995; Roberts, 2007).

Viewed through the dynamic capabilities lens (Teece, 2018; Wong & Ngai, 2025), the integration of multiple AI tools represents a firm's ability to sense opportunities for technological recombination, seize them through strategic resource reconfiguration, and transform organizational routines to embed

continuous learning (Ritala et al., 2024; Stornelli et al., 2024). These insights are synthesized by Mikalef and Gupta (2021, p. 4) who define “*AI capability as the ability of a firm to select, orchestrate, and leverage its AI-specific resources.*” The authors propose eight resources that constitute an AI capability: data, technology, basic resources, technical skills, business skills, inter-departmental coordination, organizational change capacity, and risk proclivity. Firms that develop these dynamic capabilities can orchestrate complementarities among AI technologies, transforming discrete systems into cohesive digital ecosystems. The OECD/BCG/INSEAD report (2025) finds that firms deploying multiple AI technologies achieve higher productivity and innovation intensity, confirming that complementarities operate at both the data and process levels. In this view, AI adoption is not a sequence of isolated investments but a process of capability building in which technological integration and organizational adaptation reinforce one another.

With this conceptual framework, we test the following hypothesis:

Hypothesis I. AI technology adoption is associated positively with the number of remaining AI technologies.

2.3. Economies of Scope and Cross-Functional Integration

The coexistence and integration of multiple AI technologies within firms can generate economies of scope, whereby the joint use of shared infrastructures lowers marginal costs and amplifies returns across diverse business functions (Brynjolfsson et al., 2021; Teece, 2018). Once firms invest in core data architectures and interoperable systems, the incremental cost of applying AI to new functional areas declines sharply. For example, an ML algorithm developed for production forecasting can later optimize logistics operations, while text mining and NLG models can support both marketing analytics and HR recruitment (Peretz-Andersson and Torkar, 2024). Such reuse of digital assets illustrates how AI technologies enable cumulative learning and cross-functional scaling. This is expressed by Goldfarb and Trefler (2018, p. 470) as “*the fixed cost of building an AI capacity generates economies of scope. It is only worth having an AI team within a company if there are a variety of applications for them to work on...(and more importantly) ... Economies of scope in AI ... appear to be driven by economies of scope in innovation. A wider variety of potential applications generates greater incentives to invest in an AI research team, and it generates more benefits to each particular AI project due to the potential to share data across applications*”.

Empirical evidence supports this combinatorial logic. Calvino and Fontanelli (2025) demonstrate that firms utilizing multiple AI technologies exhibit a broader diffusion of AI across functions and higher innovation intensity. Similarly, Ulrich and Frank (2021) find that German SMEs integrating AI with existing digital tools achieve more efficient workflows and faster decision-making. From a socio-technical perspective, these economies of scope emerge when technical integration is accompanied by social alignment—shared knowledge, routines, and governance that allow data and insights to circulate effectively between functions (Haefner et al., 2023).

With this conceptual framework, we test the following hypothesis:

Hypothesis II. The use of AI technologies in each functional area is positively associated with the scope of the use in other functional areas within firms.

2.4. The Strategic Challenge: No One AI Fits All Firms or Functions

Despite AI's potential, many firms struggle to identify which technologies best serve their specific needs. Arguably, one can suggest that "no one AI technology adoption fits all firms or all functions." Adoption patterns are heterogeneous across sectors, reflecting differences in task structures, data intensity, resource endowments, AI readiness, and organizational architecture (Ångström et al, 2023; Davenport, 2018; Haefner et al., 2023; Peretz-Andersson et al., 2024; Ransbotham et al., 2017; Tehrani et al., 2024). For instance, NLG and speech-based tools are prevalent in knowledge-intensive services and customer engagement, whereas image recognition and robotics are more dominant in manufacturing and logistics. Peretz-Andersson et al. (2024) study found that firms gain a competitive advantage when they align AI adoption with the cognitive nature of their core tasks—classification, prediction, or communication—rather than pursuing generic automation strategies.

This diversity underscores a fundamental managerial challenge: firms must develop AI adoption strategies that are both selective and integrative. As the research by Ransbotham et al. (2020, p. 6) found, "*Merely developing a strategy for AI is not enough...but... that tying a strategy for AI to the company's overall strategy is essential.*" Successful adopters treat AI as a modular ecosystem, building complementarities between technologies and functions to maximize returns on shared data and infrastructure. The ability to orchestrate such complementarities defines the firm's AI capability maturity—a source of dynamic advantage that allows continuous learning and innovation across domains (Makarius et al., 2020; Mikalef and Gupta, 2023; Ritala et al., 2024; Teece, 2018; Wong &

Ngai, 2025). Understanding these dynamics is crucial for explaining why some firms achieve broad performance gains through AI, while others remain technologically fragmented.

Another stream of recent managerial research highlights the risks of fragmented adoption paths. Challagalla et al. (2025) demonstrate that many firms operate an excessive number of AI pilots across various functions, technologies, and business units. While experimentation is a hallmark of dynamic capabilities, excessive or poorly coordinated pilots hinder cumulative learning and prevent firms from achieving the integration needed to unlock complementarities. Their findings reinforce the idea that AI maturity depends not on the number of pilots launched, but on the ability to scale a smaller subset of strategically relevant technologies and embed them into core workflows. This organizational capability—deciding where to experiment, how much, and when to scale—directly intersects with the technological complementarities revealed by the co-occurrence analysis in this paper (Ameye et al., 2023).

Ultimately, combining STST and dynamic capabilities theory provides a comprehensive theoretical foundation for this study. STST explains how social and technical systems interact to enable AI integration, while dynamic capabilities clarify how firms sense, coordinate, and transform these integrations over time. This joint framework directly informs the research questions posed above.

Together, these perspectives highlight that the value of AI emerges not from isolated adoption but from the ability to integrate, coordinate, and learn across both technological and organizational dimensions.

3.- Data Sources and Summary Statistics

This paper relies on firm-level data collected by the Regional Center for Studies on the Development of the Information Society, CETIC. CETIC, established in 2005, is responsible for monitoring the adoption of information and communication technologies (ICT) in Brazil. From 2005, CETIC has conducted an ICT survey to measure the ownership and use of ICT among Brazilian companies. The ICT survey adheres to the methodological standards of the Organization for Economic Cooperation and Development (OECD) and the European Commission's Statistical Office (Eurostat), thereby ensuring comparability of its results with those of OECD countries. The universe covered in the survey comprises all active Brazilian companies with ten or more employees registered in the Central Business Registry (Cempre) of the Brazilian Institute of Geography and Statistics (IBGE), belonging to the CNAE 2.0 sectors. The process of selecting firms is designed to ensure representativeness

across three domains. First, the regions that correspond to the regional division of Brazil, according to IBGE criteria, are the Central-West, Northeast, North, Southeast, and South macro-regions. Second, firms size division into small, medium, and large companies according to the number of employees, respectively, as follows: 10 to 49 employees, 50 to 249 employees, and 250 or more employees. Lastly, the market firms operate according to the CNAE 2.0 codes. The selection process is stratified, and companies are randomly selected within each domain.

Since 2019, the survey has included a section called "Novas tecnologias," which comprises, among other questions, inquiries about the adoption by Brazilian firms of big data, robots, 3D printing, and cloud computing. However, only from the surveys of 2021, 2023, and 2024, the ICT surveys included questions about artificial intelligence, specifically the AI applications or tools used, as well as the functional area of firms where AI was utilized. The inclusion of these questions enables us to examine the extent to which firms are using these AI technologies, and where within the firms they are applied, and to test the hypotheses.

For the specific AI technologies used by Brazilian firms, the ICT survey asks: Does this company use Artificial Intelligence technologies for:

- Text mining and written language analysis?
- Speech recognition converts spoken language into a machine-readable format.
- Natural Language Generation (NLG) for written or spoken language?
- Image recognition and processing, which identify objects or people?
- Machine learning, such as deep learning, for prediction and data analysis?
- Automation of workflow processes?
- Physical movement of machines through autonomous decisions, such as robots, autonomous vehicles, and drones?

For each of these AI technologies, we create a binary variable that takes the value of one if firm responded yes, and zero otherwise.

For the use of AI technologies in various functional areas of the firm, the ICT survey asks:

- Marketing or sales?
- Production processes?
- Organizing business administration processes?
- Business management?

- Logistics?
- Digital security?
- Human resources management or recruitment?

For each of these AI technologies, we create a binary variable that takes the value of one if firm responded yes, and zero otherwise.

The ICT surveys include a rich set of other ICT technologies, some of which researchers suggest proxy the degree of digitization of firms (Calvino and Fontanelli, 2025; McElheran et al., 2023; Zolas et al, 2021). Over the three years, the ICT survey included questions about the use of cloud services, Customer Relationship Management (CRM) software, and Enterprise Resource Planning (ERP) software. Following Calvino and Fontanelli (2025), we count the number of these three technologies and construct the variable "other non-AI digital technologies". Furthermore, we create a dummy variable equal to one if the firm has a fast broadband connection, i.e., a connection speed of 100 Mbit/second or greater. Unfortunately, questions on the use of e-commerce and whether a firm provided training to their employees were dropped for the 2021, 2023, and 2024 surveys. Likewise, the ICT survey does not include questions regarding firm age. Table 1 presents the definitions of variables used in the empirical exercises.

Table 1 Variables for empirical exercises

Firm characteristics	Variable
Size category	Small=20-49, Medium=50-249, Large>=+250
Fast broadband	Dummy=1 if the firm has a broadband internet connection with speed equal to or greater than 100Mbit/s, and zero otherwise
ERP	Dummy=1 if the firm uses ERP systems
CRM	Dummy=1 if the firm uses CRM systems
CC	Dummy=1 if the firm uses any of the four cloud computing paid services
Non-AI digital technologies	Sum over the use of ERP, CRM, and Cloud computing
Other AI-technologies	Sum over the remaining AI technologies except one
Other Business-functions	Sum over the remaining business functions except one

Source: Author's calculation based on ICT survey - CETIC

Table 2 reports the averages of the use of the ERP, CRM, cloud computing (CC), and the mean of the digital non-AI technologies across several dimensions. Evidently, AI users outperform non-AI users in all non-AI technologies and across all dimensions, providing a first glimpse of the interdependencies of digital technology use asserted by Zolas et al. (2021). It is worthwhile highlighting the greater use of all the non-AI digital technologies as firms grow.

Table 2. AI user and Non-AI users

All firms	AI users	Non-AI users
Other digital non-AI technologies	2.23	1.39
Fast broadband	0.832	0.694
ERP	0.719	0.375
CRM	0.613	0.276
CC	0.953	0.771
Small		
Other digital non-AI technologies	1.482	1.459
Fast broadband	0.715	0.708
ERP	0.394	0.398
CRM	0.335	0.307
CC	0.789	0.788
Medium		
Other digital non-AI technologies	2.010	1.434
Fast broadband	0.773	0.7
ERP	0.647	0.394
CRM	0.502	0.292
CC	0.914	0.781
Large		
Other digital non-AI technologies	2.140	1.464
Fast broadband	0.828	0.706
ERP	0.715	0.411
CRM	0.549	0.303
CC	0.934	0.785

Source: ICT Survey - CETIC

3.1 The use of different AI technologies within firms

The diffusion of AI in Brazil in recent years has been relatively high compared to that of European countries and has remained stable. In 2021, 19% of surveyed firms adopted at least one of the seven AI technologies. This percentage increased one percentage point to 20.0% in 2023 and decreased slightly to 19.7% in 2024.

The analysis begins by presenting the share of firms that utilized the seven AI applications identified in the ICT survey. The number of AI technologies refers to the number of "Yes" answers to the

screening question presented above. Figure 1 reports this, where several facts can be observed. First, as noted in the study by Calvino and Fontanelli (2025), there is a high dispersion in the use of different AI applications. Two applications are more widely used. These involve automating workflow processes and image recognition. The first tool, namely AI workflow automation, utilizes artificial intelligence to streamline business processes and tasks across various systems and departments. Firms using this AI tool will reduce the time spent on routine tasks, and by minimizing the need for human intervention in repetitive processes, employees can focus on higher-value work that requires more creativity and strategic thinking. Given the nature of this AI tool, its adoption may reflect its greater applicability within firms' areas. Similarly, image recognition tools or systems help make digital image and video data processing and analysis capabilities faster, more accurate, and more efficient. Second, the share of firms that adopted one of the other six AI applications is relatively similar, 3%. In general, the adoption of most AI technologies does not exhibit a clear specialization pattern.

Place Figure 1 here

Figure 2 reports the adoption of any of the AI technologies across seven functional areas of firms. There are two areas within firms where AI applications are used slightly more intensively: digital security and marketing. Furthermore, the Figure shows that AI technologies are used in all the management, HRM, and business administration functions, which is consistent with the greater use of AI tools in automating workflow processes. In some ways, the results for Brazilian firms resemble the findings for French firms in that *"AI systems may be relatively more useful in intangible, rather than physical, applications (e.g., decision support systems)"* (Calvino and Fontanelli, 2025, p. 13). Generally, there is no clear differentiation in the use of AI applications across various functional areas.

Place Figure 2 here

Figure 3 presents the share of firms that adopted any of the AI applications across the eight economic sectors defined by the CETIC- ICT survey. The patterns of AI adoption differ across industries. Not surprisingly, the Information and Communication sector leads the use of AI. About 38% of firms in the sector reported having used AI tools. This result is in line with early insights provided by Bresnahan (Bresnahan, 2022; Gerbert et al., 2017; Ransbotham et al., 2017) that in the information technology sector, companies were the leaders in achieving a substantial productivity benefit from using AI applications. In second place is the Professional, Scientific, and Technical industry, where

24% of firms used at least one AI application. Notice that these two sectors in the same order of AI use are the two highest users of AI tools in French firms (Calvino and Fontanelli, 2025).

Place Figure 3 here

How is the firms' adoption of specific AI applications across the eight Brazilian industry sectors? Moreover, how do these sectors utilize these applications within the functional areas of the firms? Figures 4 and 5 provide answers.

First, Figure 4 shows that the results for Brazilian firms reveal a distinctive and comparatively *less* specialized pattern of AI technology adoption across industries. While automation of workflow processes and image recognition emerge as the most widely adopted AI applications, their diffusion across almost all sectors suggests that Brazilian firms primarily leverage AI to optimize existing operations rather than to develop new, sector-specific technological competencies. The particularly intense adoption of automation and workflow-related AI in the Information and Communication industry aligns with this interpretation: firms in this sector are typically more digitally mature, better equipped with ICT infrastructure, and possess stronger absorptive capacities for integrating AI into routine processes (Acemoglu et al., 2025; Calvino & Fontanelli, 2023; Calvino et al., 2024).

At the same time, the absence of explicit sectoral specialization—in contrast to patterns observed in advanced economies—indicates that AI diffusion in Brazil seems to follow a relatively horizontal adoption trajectory. Firms adopt AI as a general-purpose digital tool rather than as a specialized technology tied to industry-specific applications. This is consistent with findings for other middle-income economies, where limited complementary assets, smaller firm sizes, and weaker knowledge diffusion channels constrain the development of AI-intensive innovation systems (Calvino and Fontanelli, 2023; Calvino et al., 2024; Kinkel et al., 2022).

Place Figure 4 here

From a functional perspective, Figure 5 shows that the firm and consistent use of AI in digital security reflects an approach to guard firms from cyberattacks (Singh et al., 2020). Firms prioritize AI to safeguard digital assets and mitigate risks associated with increasing online operations and transactions. This aligns with a broader Latin American trend, where cybersecurity and compliance are leading drivers of AI adoption (Cremer et al., 2022; Elsayed and Zulkernine, 2020; Gatica-Neira et al., 2023; Paramesha et al., 2024). The diffusion of AI into management, administration, and human

resources functions suggests its role as an enabler of administrative modernization, rather than as a tool for product or process innovation. Conversely, in manufacturing and logistics, AI adoption focuses on tangible operations, such as production scheduling, predictive maintenance, and workflow automation, signaling an incremental rather than transformative technological transition.

Place Figure 5 here

Given the similarity between the Brazilian ICT survey questions regarding the use of different AI applications and their functional areas of use and those of the Calvino and Fontanelli study, it is worthwhile to compare the results. The Brazilian evidence diverges markedly from the patterns observed by Calvino and Fontanelli (2025) for French firms, who report strong sectoral and technological specialization. In France, firms in ICT and Professional & Scientific services concentrate on natural language processing, machine learning, and data-driven decision-making. At the same time, manufacturing industries are increasingly relying on automation and robotics. This structure reveals technological complementarities and hierarchical dependencies—with machine learning and automation acting as foundational technologies that enable other AI systems.

In Brazil, by contrast, the technological map appears flatter and more fragmented. AI applications coexist without forming strong interdependencies, suggesting that firms deploy standalone systems tailored to immediate operational needs. This could stem from the scarcity of R&D collaboration, weak inter-firm knowledge flows (both national and international), and the limited depth of AI ecosystems (Burström et al., 2021; Cathles et al., 2022; Farrokhi et al., 2024). In such settings, AI adoption tends to be driven by the use of external, generic solutions rather than internal innovation capabilities, reinforcing the "imported technology" pattern observed in digital transformation studies of emerging economies (Verhoogen, 2023; Vujanović et al., 2022).

While French firms use AI for customer interaction, marketing, and knowledge-intensive decision-making, Brazilian firms emphasize digital security and administrative processes, reflecting different stages of digital maturity. The Brazilian pattern is consistent with a "foundational stage" of AI diffusion, where firms aim to enhance their digital infrastructure and data management capabilities. Finally, the Brazilian data show a notable exception in logistics, where AI adoption is widespread across sectors—contrary to the French case. This likely reflects the structural characteristics of the Brazilian economy, where long transportation distances, fragmented supply chains, and high logistics costs create strong incentives for AI-enabled optimization tools (Cathles et al., 2022). Such sectoral contrasts highlight how structural and institutional features, including infrastructure bottlenecks, a

lack of infrastructure sharing, and regulatory gaps, shape the technological trajectories of AI diffusion in emerging economies (Cathles et al., 2022; Katz et al., 2023).

Place Figure 6 here

Figures 6 and 7 provide further evidence on the diffusion of AI use among adopting firms, distinguishing between the number of technologies implemented and the range of business functions supported. Figure 6 reveals that AI adoption among Brazilian firms remains highly concentrated, with nearly 58 percent of adopters relying on only one AI technology and close to 9.1 percent using no more than three. This concentration indicates that, even among adopters, experimentation and implementation tend to be narrow in technological scope. Figure 7 extends this picture to the functional dimension, showing that about one-fourth (24.4%) of AI adopters employ these tools in a single business function. In contrast, a small but large fraction (8.2%) applies them across all seven areas of firm activity. This result does not align well with the average percentage of firms reporting the use of AI tools in at least one business function (53% for the years 2021-2024) in McKinsey's 2025 State of AI survey (Singla et al., 2025). This distribution suggests that while AI diffusion across firms remains limited, a small subset of firms is achieving a deeper integration of AI across organizational functions. Altogether, these results indicate an uneven pattern of AI maturity: most firms remain at an early adoption stage—testing or deploying specific tools in isolated domains—while a smaller group, likely more digitally capable or larger firms, are advancing toward more systemic and cross-functional AI integration (Brynjolfsson et al., 2025).

Place Figure 7 here

Finally, when comparing the AI adoption rate of Brazilian firms (19.5%) with those reported for developed economies, the evolution fares relatively well. In the United States, adoption rates vary considerably across studies and survey years. Zolas et al. (2020) documented a mean adoption rate of 6.6%, while McElheran et al. (2024), analyzing US firms in 2017, found a similar rate of 5.8%. In this last study, when disaggregated by firm size categories, the adoption rates are 2.7% among micro firms, 3.1% among small firms, 4% among medium-sized firms, and over 10.5% among large firms. More recently, Bonney et al. (2024), using the Business Trends and Outlook Survey (BTOS), estimated that AI use among firms rose from 3.7% in September 2023 to 5.4% by February 2024. According to McElheran et al. (2025), AI adoption increased substantially to 22.8% by 2021. Such variation across US estimates reflects differences in survey scope, firm coverage, and definitions of AI adoption. In the United Kingdom, Evans and Heimann (2022) reported that 15% of small firms

had adopted at least one type of AI technology, compared to 34% of medium-sized firms and 68% of large firms. Taken together, these figures underscore the growing diffusion of artificial intelligence—a technology increasingly regarded as a General-Purpose Technology (GPT) (Brynjolfsson & McAfee, 2011; McElheran et al., 2025). Compared to German firms (See footnote 6), the rate of adoption of Brazilian firms is relatively similar. In general, studies, such as Kasakova et al. (2020) and Ameye et al. (2023), that gather information about specific AI technology use, report higher rates of AI use. It also helps explain the somewhat high rate of AI use in Brazil.

4. Empirical Strategy

The objective of the empirical strategy is to test the hypotheses discussed in Section 2 following the econometric approach used by Calvino and Fontanelli (2025). The analysis employs regression models, presented in equation (1) below, that consider the relationship between the use of AI technologies and firm characteristics on one hand, and the use of AI applications in the seven business functions on the other hand.

4.1 Dependent variable

We follow Calvino and Fontanelli (2025) and utilize two dependent binary variables. The first is related to the use of any of the seven AI technologies firms use in their business activities. The second is the functional area where firms use any AI technology.

4.2 Independent variable

Non-AI digital technologies. This variable is constructed by summing the positive responses to a binary question about whether the firm used ERP, CRM, and cloud computing. This variable is a proxy of the level of digitalization the firm has.

Fast broadband. This binary variable takes the value of one if the firm has a broadband internet connection with a speed of 100 Mbps or greater, and zero otherwise.

We use two different independent variables of interest in equation (1) below.

AI-Technologies. Our first variable of interest is the count of six out of seven AI technologies that firms use in their business processes for the production or development of products and services (Calvino and Fontanelli, 2025). The seven AI technologies defined in Section Three are: natural

language generation (NLG), image recognition, machine learning, workflow process automation, and robotics. When one AI technology is the dependent variable, the count is over the six remaining AI technologies.

Functional areas using AI. Our second variable of interest is the count of six out of seven business processes for the production or development of products and services where firms use any of the AI technologies. These areas include marketing, production, business administration, management, logistics, digital security, and human resources management (HRM for short). When the dependent variable is one of the functional areas where AI is utilized with a specific AI technology, we create a variable that aggregates the six remaining functional areas where AI is employed.

4.3 Control variable

The ICT survey by CETIC categorizes sizes into three categories: small, medium, and large. In all regressions, we control for size, taking the small group as the reference. As the Brazilian ICT survey is a cross-sectional dataset, we control for industry at the one-digit level, region, and year fixed effects.

In this study, we use the following model to test hypotheses 1 and 2.:

$$AI_{i,t}^j = \alpha + B_1 Size_{i,t} + \beta_2 Fast\ Broadband_{i,t} + \beta_3 Non - AI\ digital_{i,t} + \beta_4 \sum Other\ AI_{k \neq i,t} + FE_{s,t,r} + \epsilon_{i,t} \quad (1)$$

To make clear the equation, the primary variable of interest, Other AI, is the count of other AI technologies (when i refers to a technology) or business functions (when j , the superscript, refers to a business function) adopted by the firm, excluding the dependent variable (i or j). Last, unfortunately, the ICT survey by CETIC does not include firm age, a variable commonly included in all the studies that address the determinants of digital technologies adoption (Calvino and Fontanelli, 2025; Cho et al., 2023; DeStefano et al., 2023; McElheran et al., 2024; Zolas et al., 2021). While important, the inclusion of other firm characteristics provides sufficient support for the models.

5. Results

5.1 Interdependence or complementarity among AI technologies.

Table 3 examines the relationship between firm size, fast broadband access, and non-AI digital technologies and the adoption of various AI applications. Using small firms as the reference group,

the results reveal distinct size patterns once other AI tools are controlled for. For image recognition, workflow automation, and machine learning, AI-adopting firms are, on average, larger than non-users, consistent with Calvino and Fontanelli (2025). Conversely, the use of text mining and NLG shows negative associations with firm size, suggesting that smaller or more flexible firms may be more inclined to adopt these tools. The evidence is mixed for speech recognition and robotics: large firms appear less likely to adopt speech recognition, while robot adoption remains concentrated among large or very large firms, reflecting the need for higher capital and complementary IT capabilities.

The results also underscore the importance of a firm's digital maturity in shaping AI adoption. The positive and significant associations between non-AI digital technologies (ERP, CRM, cloud) and the adoption of speech recognition, NLG, machine learning, and automation support Ameye et al., (2023), and Calvino and Fontanelli (2025) and Zolas et al's. (2021), argument that digitally advanced firms are better positioned to use data-intensive AI tools. In contrast, the weak or negative associations of AI technologies and firms' digitization for image recognition and robotics suggest that AI linked to perceptual or physical tasks depends more on hardware and sensor integration than on business software. Interestingly, unlike Calvino and Fontanelli, this study finds that NLG adoption is positively associated with non-AI digital technologies, indicating that among Brazilian firms, AI tools are becoming embedded in existing systems such as ERP-based document automation or CRM-driven content generation.

Finally, broadband availability does not emerge as a systematic driver of AI technologies adoption. Fast broadband shows no significant association with most AI technologies once a basic connectivity threshold, i.e., 100 Mbps or higher, is met. However, the positive coefficient for machine learning and the negative one for robotics suggest nuanced effects: broadband likely facilitates cloud-based, data-intensive ML applications, whereas robotics depends on local processing and physical integration. Overall, for Brazilian firms, broadband appears to be an enabling but not decisive factor—AI adoption is ultimately driven more by internal digital capabilities than by external network infrastructure.

Table 3. AI technologies and their interconnectedness

Text Mining	Speech Recognition	NLG	Image Recognition	ML	Automation	Robots
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Medium	-0.300** (0.120)	0.109 (0.080)	-0.195* (0.117)	0.165** (0.078)	0.209*** (0.075)	0.209*** (0.037)	-0.021 (0.133)
Large	-0.305*** (0.105)	-0.126* (0.072)	-0.262*** (0.080)	0.421*** (0.102)	0.184*** (0.070)	0.294*** (0.059)	0.181** (0.074)
Fast broadband	0.066 (0.115)	0.025 (0.084)	0.122 (0.114)	0.060 (0.093)	0.207*** (0.080)	-0.005 (0.015)	-0.097* (0.054)
Non-AI digital technologies	0.072 (0.055)	0.108** (0.049)	0.118*** (0.017)	-0.043* (0.022)	0.251*** (0.055)	0.228*** (0.012)	0.007 (0.020)
Other AI	0.875*** (0.029)	0.790*** (0.022)	0.825*** (0.027)	0.847*** (0.034)	0.783*** (0.003)	1.414*** (0.103)	0.588*** (0.029)
Constant	-3.066*** (0.147)	-3.067*** (0.098)	-3.240*** (0.111)	-1.888*** (0.072)	-2.888*** (0.146)	-3.197*** (0.136)	-2.223*** (0.055)
Observations	12675	12605	12802	12903	12556	12979	12246
Pseudo R2	0.575	0.571	0.587	0.463	0.536	0.588	0.436
Industry-Region-year FE	yes	yes	yes	yes	yes	yes	yes

Industry-Year-Region FE are jointly defined for industry, region, and year. Robust standard errors in clustered at the 1-digit industry level in parentheses

The results for the other AI technologies variable reveal a strong and highly significant positive association across all models, indicating that firms adopting one AI technology are substantially more likely to implement others. This result answers research question 1 and supports Hypothesis I. This finding closely mirrors Calvino and Fontanelli's (2025) evidence for French firms, reinforcing the notion that AI applications tend to be adopted in complementary bundles rather than as isolated tools. Such complementarities likely emerge because the capabilities required to implement one AI technology—such as data management, algorithmic literacy, or digital infrastructure—can be leveraged to deploy others, reducing marginal adoption costs and fostering technological synergies. This pattern underscores the combinatorial nature of AI innovation, where the integration of multiple AI components (e.g., machine learning, natural language generation, and automation) amplifies functionality and business value. It also aligns with the broader literature on cumulative technological capabilities (Andrews et al., 2018; Ciarli et al., 2021; Greenan and Napolitano, 2024), which suggests that firms build dynamic complementarities between digital technologies over time. For Brazilian firms, this interdependence implies that AI diffusion is likely to proceed through clusters of related technologies, with early adopters deepening and diversifying their AI portfolios as organizational and data capabilities mature. We expand this analysis in the last sub-section.

5.2 Interdependence in the use of these AI technologies across functional areas of firms

Table 4 addresses question 2, examining the association between size, non-AI digital technologies, and other areas of AI adoption within firms. Consistent with Calvino and Fontanelli's (2025) findings for French firms, the Brazilian results reveal that firm size plays a differentiated role across functional areas in shaping AI adoption. In both contexts, larger firms show a significantly higher likelihood of deploying AI in production processes, logistics, and digital security, reflecting the capital-intensive and infrastructural requirements of these applications. These functions often entail integration with physical assets, automation systems, or cybersecurity infrastructures, implying higher fixed costs and greater technological complexity—barriers more easily overcome by large firms with substantial resources. However, in contrast to the French evidence, where size effects weaken for intangible-intensive areas such as marketing or organizational processes, the Brazilian results highlight a stronger positive association of firm size with human resources and digital security, suggesting that AI-enabled HRM and cybersecurity solutions are still in early diffusion stages and concentrated among larger, digitally advanced Brazilian firms. The absence of significant size effects in business administration and management indicates that AI tools in these functions have become relatively accessible, as firms of different sizes can adopt standardized, software-based applications without requiring significant infrastructural investments. Overall, these results indicate that in emerging economies, scale remains a key enabler for adopting complex, asset-intensive AI applications. At the same time, smaller firms can participate in the AI transition primarily through standardized, software-based tools that demand lower upfront investments.

Table 4 also highlights that the diffusion of AI technologies within firms is closely linked to their digital maturity, as evidenced by the positive and significant coefficients for non-AI digital technologies across several functional areas. The results indicate that firms already using ERP, CRM, or cloud systems are more likely to integrate AI into marketing, business administration, management, and digital security activities—functions that rely heavily on information flows and structured data. This pattern supports the argument that an established digital infrastructure lowers the marginal cost of adopting AI by providing compatible data architectures and process integration (Jung and Gómez-Bengoechea, 2025; Jovanovic et al., 2022; Zolas et al., 2021). Interestingly, while Calvino and Fontanelli found such effects only for commercial and organizational processes, the Brazilian evidence suggests that digital maturity plays an enabling role across a broader range of intangible-intensive functions, possibly reflecting firms' reliance on cloud-based AI solutions. In

contrast, the absence of significant associations for production and logistics functions suggests that these activities rely more on the integration of AI with physical assets and automation systems, rather than on pre-existing software infrastructures—a pattern consistent with findings for advanced economies.

Table 4. AI technologies use in Functional areas and their interconnectedness

Variables	Marketing	Production	Business Adm	Management	Logistics	Digital Security	Human Resources
Medium	-0.168 (0.107)	0.043 (0.116)	0.054 (0.089)	-0.088 (0.120)	0.060 (0.090)	0.171*** (0.033)	0.251*** (0.068)
Large	-0.410*** (0.126)	0.113* (0.066)	0.106 (0.108)	-0.073 (0.059)	0.289* (0.168)	0.388*** (0.117)	0.489*** (0.061)
fast_broadband	0.147*** (0.051)	-0.051 (0.077)	0.040 (0.092)	0.006 (0.040)	-0.026 (0.099)	0.278*** (0.028)	-0.091 (0.112)
Non-Ai digital technologies	0.277*** (0.030)	0.043 (0.048)	0.147*** (0.030)	0.076** (0.036)	-0.018 (0.056)	0.147*** (0.028)	0.092** (0.036)
Other Functional area with AI use	0.699*** (0.011)	0.647*** (0.053)	0.792*** (0.025)	0.797*** (0.020)	0.689*** (0.030)	0.730*** (0.031)	0.722*** (0.014)
Constant	-2.650*** (0.089)	-2.796*** (0.158)	-3.033*** (0.065)	-2.845*** (0.053)	-2.324*** (0.059)	-3.558*** (0.050)	-3.014*** (0.032)
Observations	12979	12875	12923	12843	12843	12923	12923
Pseudo R2	0.655	0.659	0.741	0.741	0.693	0.662	0.706

Industry-Year-Region FE are jointly defined for industry, region and year. Robust standard errors in clustered at 1-digit industry level in parentheses. *** p<0.01, ** p<0.05, * p<0.1

The powerful and significant relationship between the use of AI in other functional areas and the likelihood of AI adoption within a given area underscores the complementary and cumulative nature of AI integration. This result provides an answer to research question 2 and supports Hypothesis II. As in Calvino and Fontanelli's (2025) study, this result suggests a reinforcing dynamic of technological interdependence: once firms develop the capabilities to apply AI in one domain, they

are better positioned to redeploy those competencies in other domains. This finding aligns with the concept of combinatorial innovation, whereby the interaction among multiple AI systems—such as analytics, automation, and natural language processing—generates increasing returns to digital adoption (Brynjolfsson and McAfee, 2011; McElheran et al., 2024). Moreover, the availability of fast broadband exhibits heterogeneous effects: while it is not a universal determinant of AI diffusion, its positive and significant coefficients for marketing and digital security suggest that in contexts with less uniform connectivity, reliable broadband remains a crucial enabling condition for data-intensive or cloud-mediated AI applications. Overall, these results highlight that for Brazilian firms, AI adoption unfolds as an incremental and cumulative process, contingent on both prior digital capabilities and the progressive deepening of AI use across interrelated business functions.

5.3 What kind of Complementarity of AI technologies?

The stylized facts in Figure 6 and the findings in Table 3 reveal two related aspects of AI use by Brazilian firms. First, a sizable number of firms (41.7%) use two or more AI technologies, and second, the adoption of a specific AI technology is enhanced with the use of the remaining six technologies. A further question that we investigate is the extent to which the seven AI technologies are complementary. To answer this question, we follow Calvino and Fontanelli (2015) and estimate the probabilities of adopting an AI application conditional upon the adoption of the remaining six AI technologies. Since we have data for the years 2021, 2023, and 2024, we perform this exercise for each year to capture some dynamics in their interdependence.

This approach measures the likelihood of adopting a specific AI technology given that another is already in use, thus identifying the direction and intensity of technological complementarities. Formally, the conditional probability represents the probability that a firm uses technology i conditional on its adoption of technology j , calculated as the ratio between the number of firms using both technologies and the total number of firms employing j . Because adoption rates differ across technologies, these probabilities are inherently asymmetric [$p_{i|j} \neq p_{j|i}$], capturing the extent to which some AI tools tend to act as foundational enablers. In contrast, others are primarily dependent applications, i.e., where some technologies act as enablers (appearing more often in rows), while others are predominantly supported technologies (appearing more often in columns). This framework allows us to trace both direct and indirect linkages across AI technologies, thereby uncovering the underlying structure of technological interdependencies within firms' digital ecosystems. The outcome is summarized in the following graphs by year, in the form of a heatmap.

Distinctly from Calvino and Fontanelli, our analysis will focus on the dynamics of change in AI technology co-occurrence (2021–2024). First, the three cross-sectional snapshots (2021, 2023, and 2024) collectively reveal a non-linear evolution in the complementarities among AI technologies. While one would expect a steady diffusion pattern—where co-occurrences increase consistently over time as technologies spread and firms learn to integrate them—the results suggest a more complex trajectory. Specifically, early complementarities (2021) were highly concentrated around two central technologies (mainly machine learning and automation of workflow processes)¹It became more diversified in 2023 with the growing involvement of language and perception-based AI (e.g., speech recognition, NLG, and image recognition). By 2024, the system exhibited signs of both diffusion and differentiation.

In 2021, co-occurrence probabilities were highest between automation and machine learning (0.71), speech recognition (0.70), and NLG (0.69). These strong complementarities reflected early-stage adoption dynamics, likely concentrated among digitally advanced firms capable of combining analytical (ML) and operational (automation) technologies. The ecosystem at that point was hierarchical and efficiency-driven, dominated by firms seeking productivity improvements through data-driven process automation.

By 2023², the pattern diversified substantially. Co-occurrences involving automation remained strong but spread across a broader range of technologies, including text mining and robots, indicating a horizontal expansion of AI integration across different organizational functions. The inclusion of text- and speech-related AI in top-ranking pairings suggests that firms have begun to embed cognitive capabilities—understanding and generating language—into their operational and customer-facing processes.

In 2024, the system became more mature yet heterogeneous. Automation continued to anchor the co-occurrence structure, but its strongest linkages slightly declined in magnitude (e.g., ML–Automation dropped from 0.71 in 2021 to 0.69 in 2024). At the same time, new, mid-level complementarities—such as speech recognition–NLG (0.51) and robots–image recognition (0.49)—gained strength, suggesting a reorganization of interdependencies rather than simple growth or decline. This shift indicates that as adoption broadened to a broader base of firms, the relative intensity of co-adoption

¹ McKinsey (2022) presented a review of AI use from 2018 to 2022, the years during which the company initiated a survey on AI use. In that report, the use of robotic process automation (Close to the automation of workflow processes in the Brazilian ICT survey), computer vision, and NLG were the most used AI tools.

² 2023 is the year that generative AI made its debut. No data on this type of AI appears in the survey.

among the most advanced pairs diluted, while new functional pairings emerged. This phenomenon—broad diffusion with compositional turnover—is typical when technologies transition from being used by leaders to being adopted by mainstream users.

Place Figure 8 here

Place Figure 9 here

Place Figure 10 here

How to interpret these dynamics?

One key factor to consider is that, since the data are cross-sectional, the observed changes may reflect compositional shifts in the population of adopters, rather than the longitudinal behavior of the same firms. Therefore, changes in co-occurrence patterns may result from four factors. First, the entry of new adopters. These firms, being potentially less digitally mature, begin adopting AI by starting with single or simpler technologies (e.g., automation alone) rather than sophisticated multi-technology integrations. This entry effect can lower average co-occurrence probabilities even if the number of adopters rises. Second, there can be a sectoral reweighting, i.e., adoption expands more quickly in specific sectors (e.g., services adopting NLP and automation), which alters the aggregate structure of complementarities. Thus, what appears as a decline in one pair (e.g., ML–automation) may actually reflect sectoral shifts toward other combinations that are more relevant to new adopters. Third, a process of technological maturity and standardization can exist. As AI tools become easier to deploy, firms can integrate them independently rather than relying on tight coupling with other systems. This leads to a decentralization of complementarities—the technologies diffuse more widely, but their joint adoption becomes more context-specific. Lastly, some form of organizational learning and experimentation may have occurred. Over time, firms may reconfigure their AI portfolios, abandoning some combinations that proved less synergistic while strengthening others that align better with evolving business models (e.g., combining speech recognition with NLG for customer service instead of with ML for analytics).

5. 4 AI Technologies Across Business Functions: Patterns of Application and Integration

The last question we want to answer is how firms utilize the seven AI applications across the seven functional areas of a firm. To answer this question, as in the previous subsection, we estimate the

likelihood of adopting a specific AI technology, conditional on the functional area in which it is used, and present the outcomes using heatmaps.

Place Figure 11 here

Place Figure 12 here

Place Figure 13 here

The conditional probability matrices for Brazilian firms, Figures 11, 12, and 13 reveal the following configuration of AI-business function interdependencies.

Across the three cross-sectional waves, a clear reorganization of complementarities emerges. For the year 2021, the co-occurrence structure was highly concentrated in marketing and digital security, dominated by speech recognition, NLG, and text mining. These early complementarities reflected communication-intensive and data-processing functions—low-hanging fruits for AI adoption. In 2023, the complementarities seem to have intensified and diversified. High conditional probabilities were observed not only in marketing and digital security, but also in HRM and business administration (e.g., robots – HRM = 0.68, robots – business administration = 0.60). This suggests that AI has moved from analytics and communication tasks toward operational and cognitive automation of workflow processes. However, in 2024, the previous pattern began to flatten partially. Conditional probabilities became more homogeneous across functions, clustering around 0.45–0.55, with fewer extreme highs. Marketing remained an anchor, but machine learning, speech recognition, and robotics appeared across broader functional domains, suggesting diffusion and normalization rather than concentration.

How to interpret this pattern?

First, again, we emphasize that the data are cross-sectional, and changes do not reflect within-firm evolution but rather composition effects and aggregate reconfiguration in the population of adopting firms across the three years. Several mechanisms could explain the observed dynamics. First, as analyzed in the previous section, there may have been entry of new adopters and compositional diffusion. Between 2021 and 2024, many new firms are likely to have entered the AI user population—especially SMEs adopting more straightforward or standardized tools. This broadening of the adopter base dilutes early high co-occurrences (since newcomers tend to start with one or two

non-integrated applications). As a result, 2024 shows flatter conditional probabilities—not because integration declined, but because the marginal adopter could have been less digitally mature.

Second, as firms used (And learned from) different AI technologies, a process of maturation and standardization of core AI tools could have occurred. Technologies such as speech recognition, NLG, and ML have become more accessible via cloud APIs and SaaS platforms. This reduces the technological threshold for adoption and facilitates entry into multiple functional areas, especially marketing, digital security, and business administration. However, as these tools commodify, their co-occurrence with others stabilizes—moving from strong “elite” complementarities to “common” coexistence patterns.

Third, as several reports suggest (Challagalla et al., 2025; Chui et al., 2018; de Bellefonds et al., 2024; McKinsey, 2024, 2025; Ransbotham et al., 2020), firms that learned from the use of a specific AI tool experimented with shifts in their strategic priorities and cost-benefit alignment. By 2023–2024, firms may realize that combining too many complex AI tools generates integration costs that exceed the benefits. The 2023 peaks (e.g., HRM–Robots = 0.68) reflect an experimental phase where firms were testing multiple applications. The subsequent moderation in 2024 suggests strategic refocusing—retaining high-value complementarities and abandoning marginal ones.

Last, a process of sectoral diffusion rather than functional diversification could have occurred. Instead of each firm using AI in more areas, i.e., intra-firm diffusion, the broader economy diversified functionally (Inter-firm diffusion): some sectors specialized in marketing AI. In contrast, others focused on automation or security. Cross-sectional data capture this shift as a redistribution rather than proper firm-level diversification. These two results capture the theoretical and empirical insights from intra- and inter-firm diffusion theory (Battisti and Stoneman, 2003, 2005; Geroski, 2000).

The dynamic patterns reported in the co-occurrence matrices align with recent evidence on how firms sequence and coordinate the adoption of AI. Ameye et al. (2023) argue that firms often navigate AI adoption under high uncertainty³ regarding both the functionality and the business relevance of

³ Ameye et al. (2023, p. 2) ask: “*uncertainty about the technology’s use cases, i.e., for what purpose should firms use AI in the first place?*” As they assert, uncertainty determines whether firms adopt or delay AI adoption: “*firms will find it more profitable to delay adoption until uncertainty gets reduced or expected returns to the technology increase enough to overcome uncertainty*” (p. 4).

different AI variants. This helps explain the shifts observed between 2021 and 2024 in Brazil, where some technologies (e.g., speech recognition, NLG, workflow automation) become central anchors of AI portfolios while others diffuse more unevenly. Similarly, Challagalla et al. (2025) explain why complementarities may strengthen or weaken over time: when firms launch many isolated pilots, co-occurrence across technologies appears fragmented, while more strategic, scaled deployments produce tighter clusters of related tools. The evolving structure of complementarities in our data likely reflects this managerial transition from pilot-heavy experimentation toward more focused capability building.

6. Discussion and Conclusions

This study provides novel evidence on how firms integrate and combine different artificial intelligence (AI) technologies, as well as on their application across various functional areas of business operations. Using unique data from Brazilian firms, we show that AI adoption within firms is characterized by significant interdependencies—both among technologies and across functional domains. Specifically, our results indicate that firms adopting one AI technology are substantially more likely to adopt others, and that the use of AI in one functional area often coexists with its application in other areas of the organization. These findings confirm that AI operates less as a set of independent tools and more as an interconnected system whose value emerges through complementarities and shared digital infrastructures (Calvino and Fontanelli, 2025; de Bellefonds et al., 2024; Gerbert et al., 2017; OECD/BCG/INSEAD, 2025).

The results also highlight the importance of firm size and digital maturity in shaping AI adoption patterns. Larger firms are more likely to adopt image recognition, machine learning, automation, and robotics, reflecting their greater capacity to invest in complex technologies and integrate them into production and logistics (Acemoglu et al., 2023, 2025). In contrast, smaller firms tend to rely on simpler, more accessible applications such as text mining or speech recognition. Fast broadband access—often considered a key digital enabler—was not a significant determinant of AI adoption except for machine learning, suggesting that connectivity alone is insufficient without complementary capabilities. More decisive was the extent of digitalization, as measured by the number of non-AI technologies used: firms with richer digital ecosystems were systematically more likely to adopt multiple AI tools (Ameye et al., 2023; DeStefano et al., 2023; Calvino and Fontanelli, 2025). Finally, the variable capturing the count of other AI technologies proved to be a strong and consistent predictor across all models, confirming the existence of technological complementarities and reinforcing the

idea that the benefits of AI adoption accumulate through combinatorial integration (Ameye et al., 2023; Calvino and Fontanelli, 2025).

The analysis of AI use across functional areas provides further insight into how these complementarities translate into organizational practice. AI adoption was most prevalent in digital security, marketing, and commercial activities, and logistics—areas where data flows are abundant, and automation yields immediate efficiency gains. In contrast, adoption in human resources and administrative management remains relatively lower, likely reflecting challenges in integrating cognitive or language-based systems with existing workflows, or some issues related to ethical factors (Andrieux et al., 2024; Tambe et al., 2019). Notably, large firms again stand out as more frequent adopters across production, logistics, and HRM, consistent with economies of scale and scope in AI deployment. Moreover, the finding that AI use in one functional area significantly increases the probability of adoption in others confirms that internal learning and reuse of AI assets drive cross-functional diffusion.

From a managerial perspective, these findings suggest that firms should view AI adoption as a strategic system-building process rather than a collection of isolated projects (Agrawal et al., 2022; Ameye et al., 2023; Leyer and Schneider, 2021; McKinsey, 2025). Managers should prioritize establishing shared data infrastructures, interoperable systems, and cross-functional teams that facilitate the reuse of AI models and insights across departments. Investment in workforce training is equally critical, not only to build technical expertise but also to cultivate organizational learning mechanisms that enable employees to identify complementarities between AI tools. In smaller firms, incremental and modular adoption strategies—starting with technologies that address specific pain points and progressively integrating others—can help overcome resource constraints while capturing economies of scope.

Our findings also speak to two other managerial and theoretical themes identified in recent research. First, the variability and evolution in co-occurrence patterns align with the uncertainty-driven herding mechanism described by Ameye et al. (2023). When firms lack explicit knowledge of which AI technologies are most suitable for specific functions, they often adopt broad categories of AI—such as "using AI"—but with significant heterogeneity in their internal configurations. This helps explain why complementarities may fluctuate over time and why no stable "dominant bundle" emerges across all firms. Second, the uneven strengthening of specific AI pairings, together with persistent fragmentation in others, reflects the organizational tendency identified by Challagalla et al. (2025):

running many pilots without scaling. AI portfolios become dispersed when firms experiment in multiple functions without developing the integration capabilities needed to embed these tools in workflows. The implication is that managing AI complementarities requires not only technological selection but also prioritization, scaling discipline, and the transformation of core processes.

Despite its contributions, this study has some limitations that open avenues for future research. First, the analysis relies on cross-sectional survey data, which constrains our ability to capture temporal dynamics and causal relationships in firms' AI adoption trajectories. Longitudinal research would help reveal how complementarities among AI technologies evolve as firms move from experimentation to large-scale implementation, and how AI use varies across functional areas within firms, i.e., the intra-firm diffusion. Second, while the Brazilian context provides valuable insights into emerging-market dynamics, institutional and infrastructural conditions may differ significantly from those in advanced economies, limiting the generalizability of the findings. Third, the study focuses on the presence or absence of AI technologies and their co-occurrence across functions, without measuring intensity, sequencing, or the specific performance impacts of integration. A final limitation is the absence of certain variables that have been identified as complementary or explanatory factors in the adoption of advanced digital technologies, with human capital being one of the most significant (Cho et al., 2022; Calvino and Fontanelli, 2022, 2023; Fontanelli et al., 2024).

Future research should therefore employ a firm-level panel to uncover the microfoundations of AI complementarities (McElheran et al., 2025)—such as managerial cognition, organizational learning routines, and digital capability development—that shape the success of systemic AI adoption. Comparative studies across countries and sectors could also help clarify how institutional contexts mediate firms' ability to exploit economies of scope from AI integration.

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No One AI Fits All: Technology and Functional Complementarities in Firm-Level AI Adoption (2021–2024)

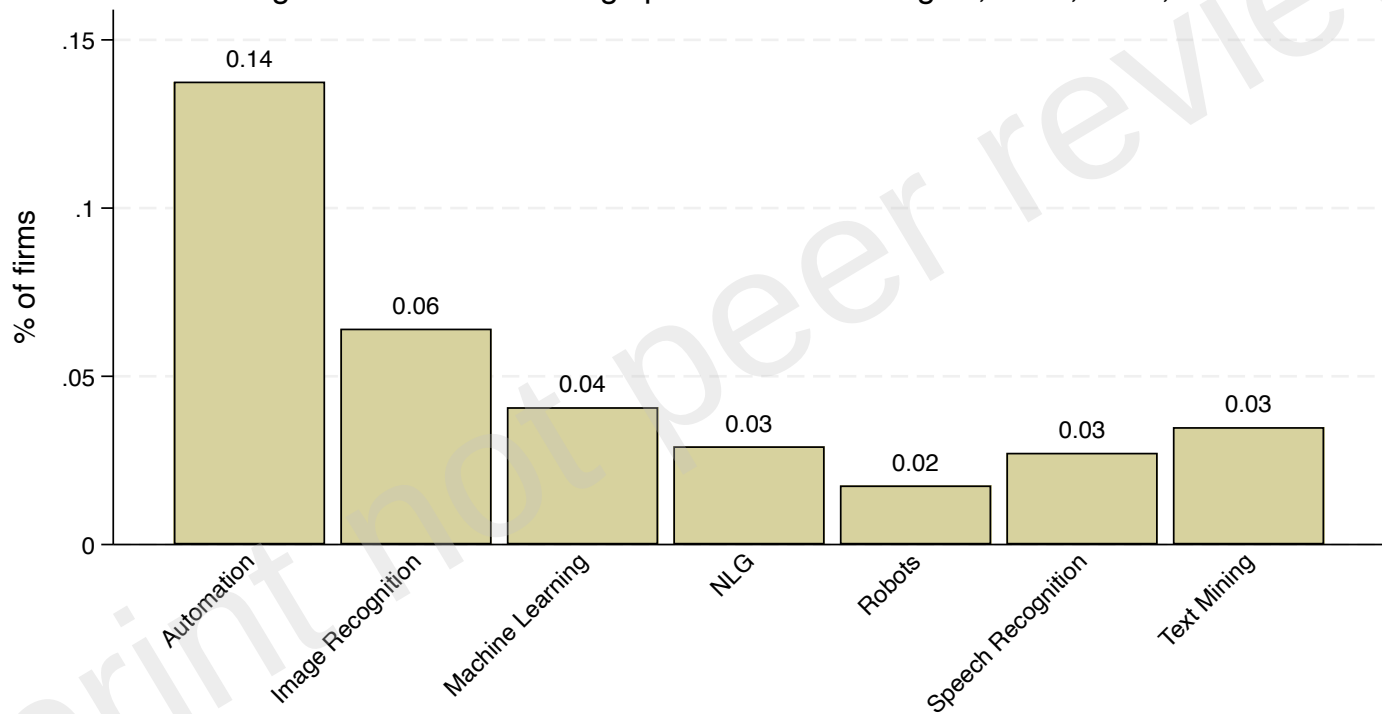
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Declaration of interest statement

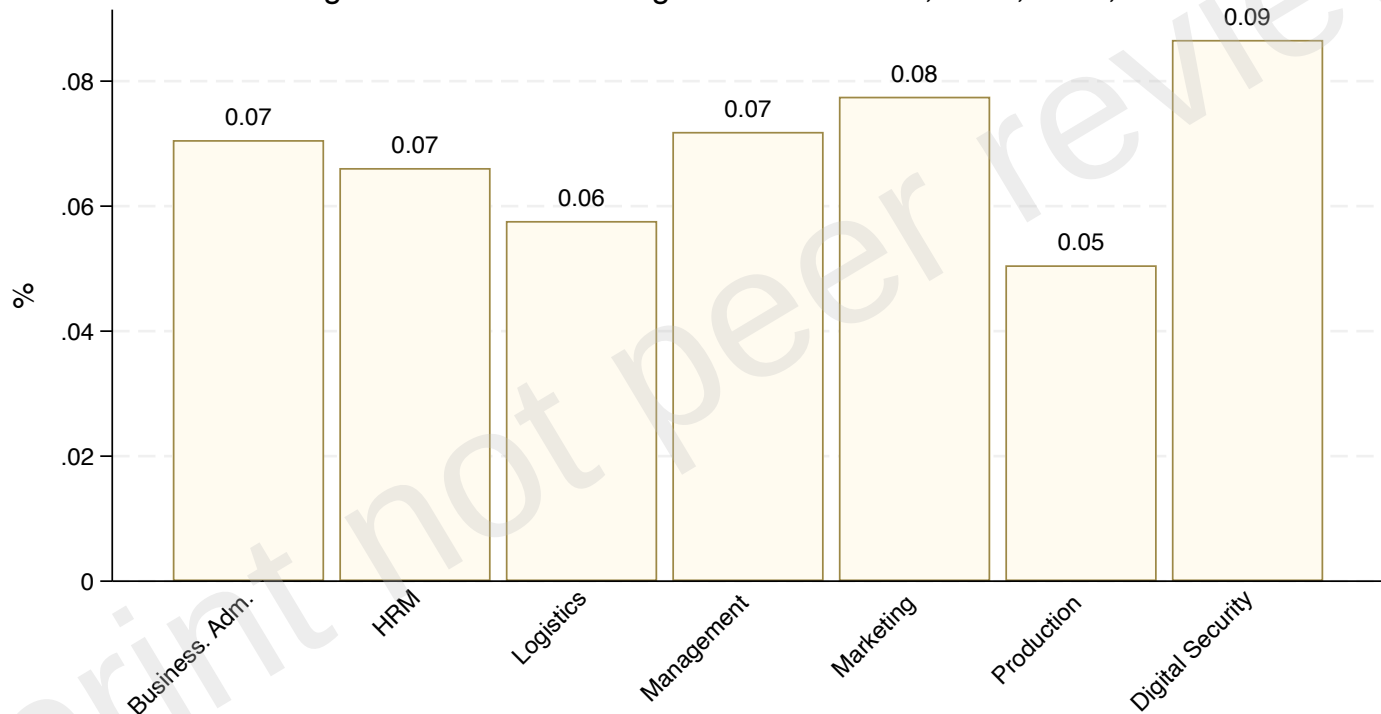
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Figure 1. % of firms using specific AI Technologies, 2021, 2023, 2024



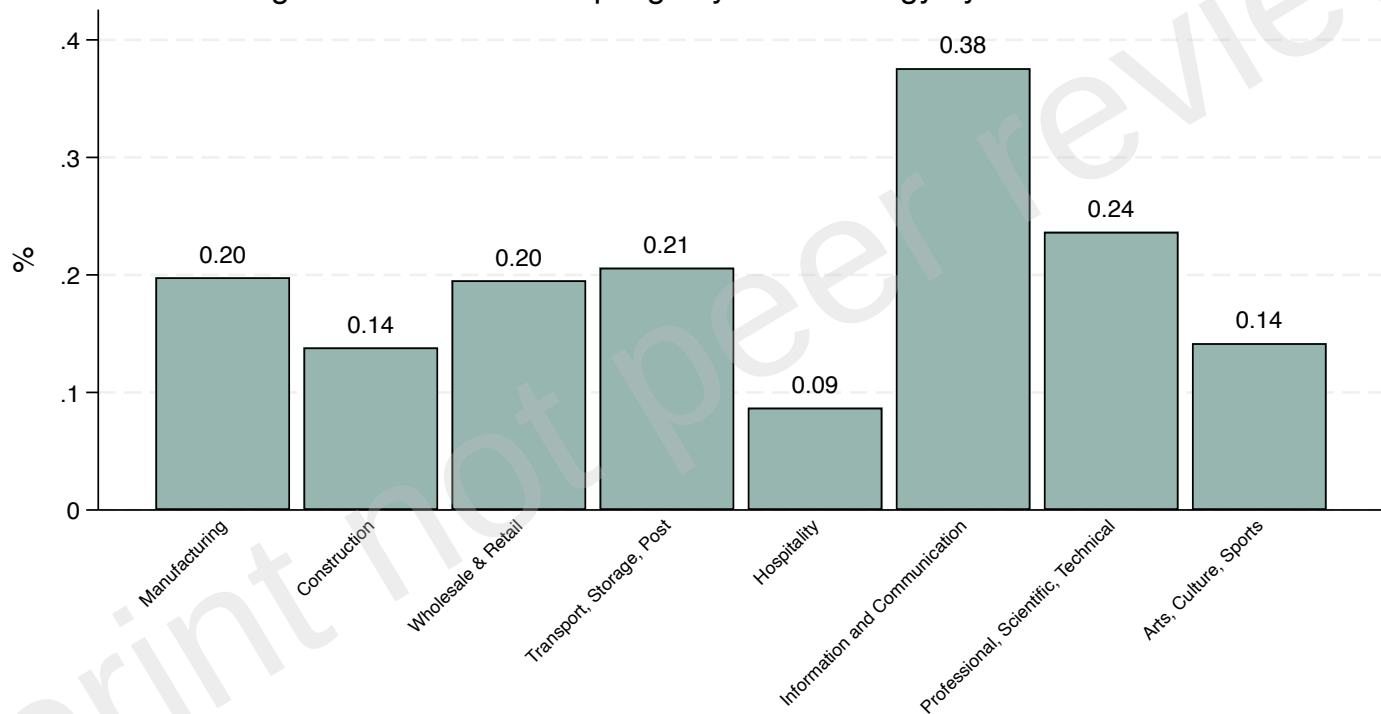
Source: ICT Survey, CETIC

Figure 2. % of firms using AI in firms' Areas, 2021, 2023, 2024



Source: ICT Survey, CETIC

Figure 3. % of firms adopting any AI technology by Economic Sector



Source: ICT Survey, CETIC

Figure 4. Sectoral share of firms using AI technologies

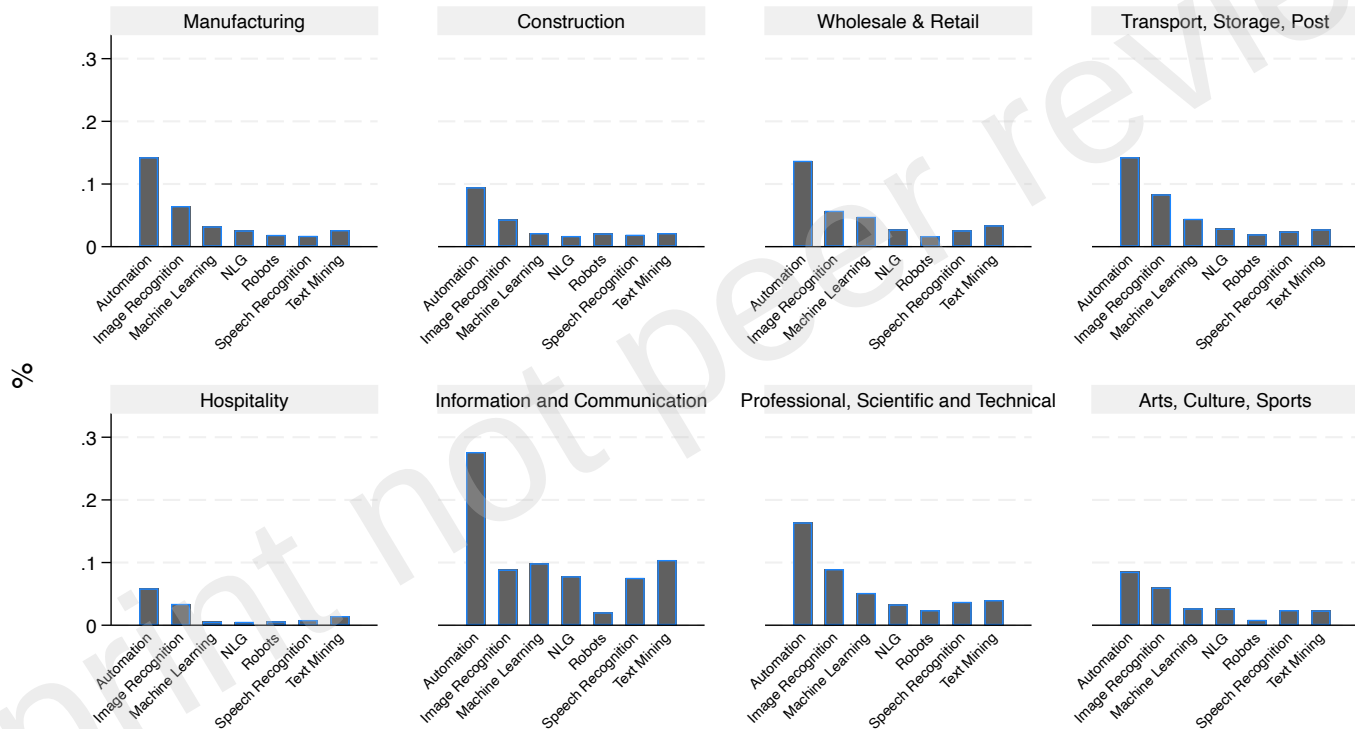


Figure 5. Sectoral share of firms using AI according to firm area of use

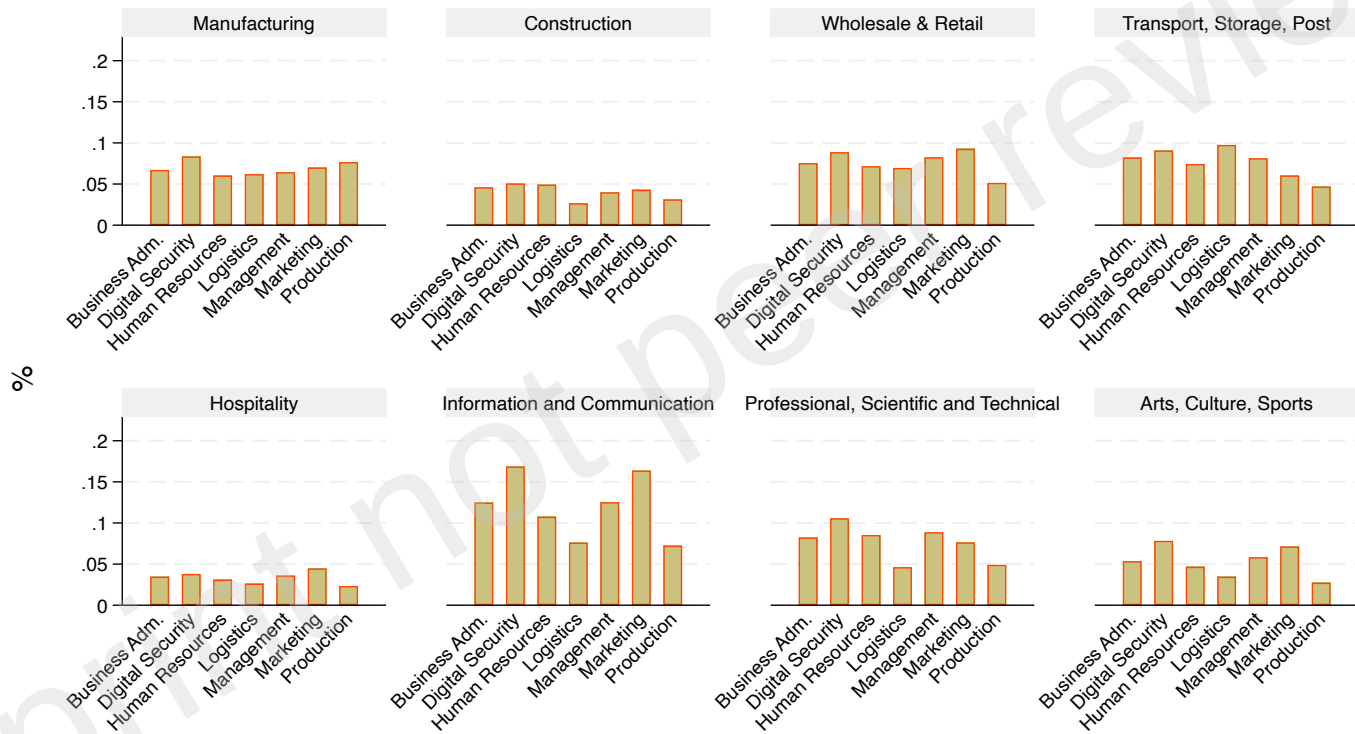
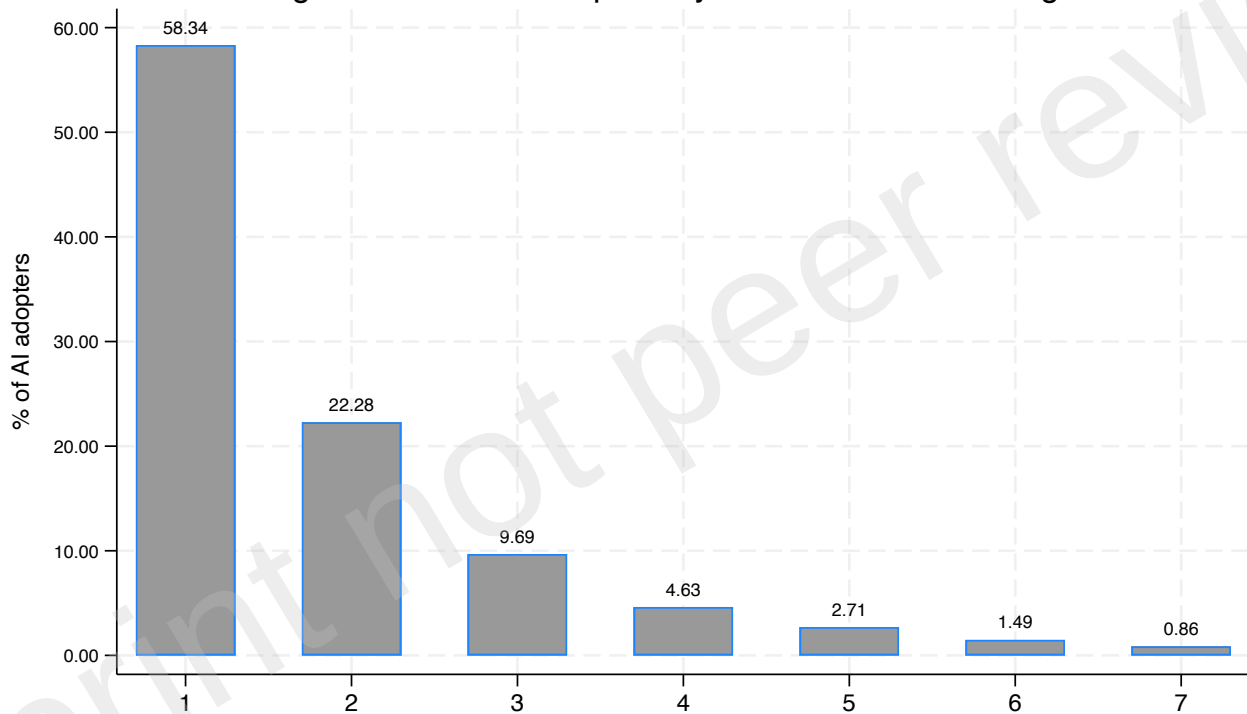


Figure 6. Share of adopters by number of AI technologies



Source: ICT Survey - CETIC

Figure 7. Share of Business functions that use at least one AI Tool



Source: ICT Survey - CETIC

Figure 8. Complementarities between the use of AI technologies

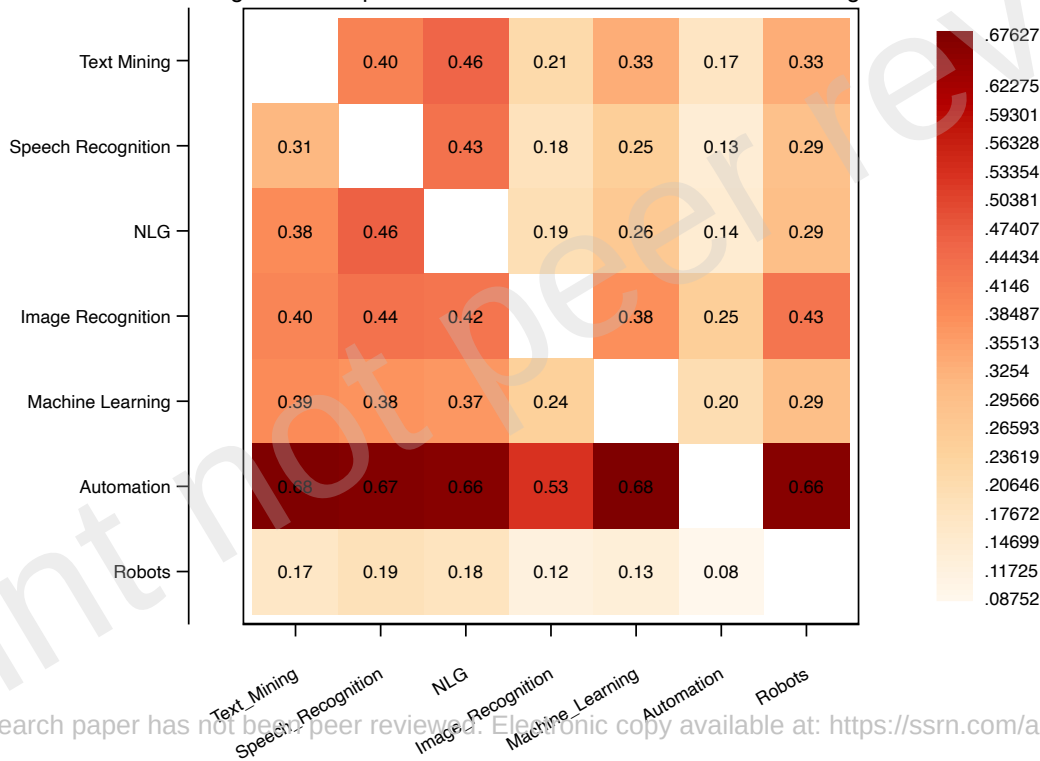


Figure 9. Linkages between AI technologies and Business Function

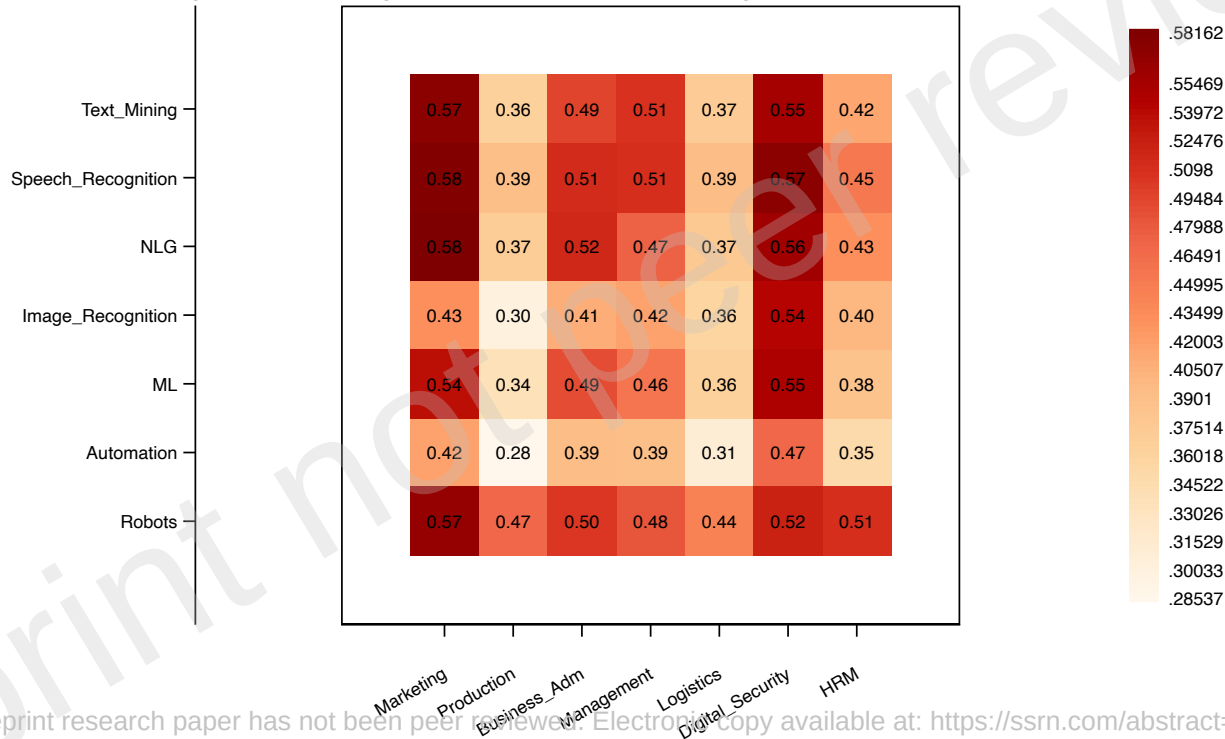


Figure 10. Complementarities between the use of AI technologies 2024

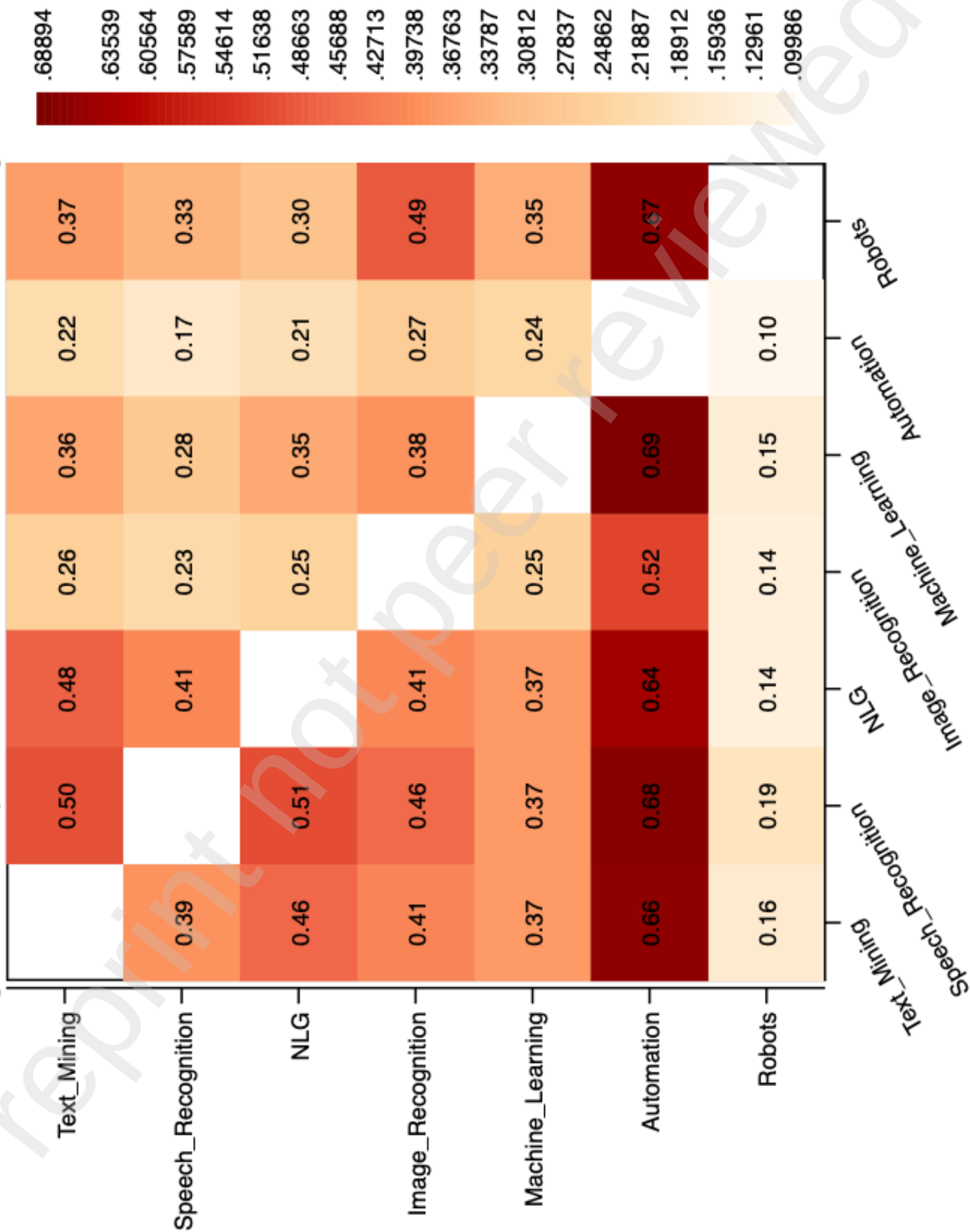


Figure 11. Linkages between AI technologies and Business Function 2021

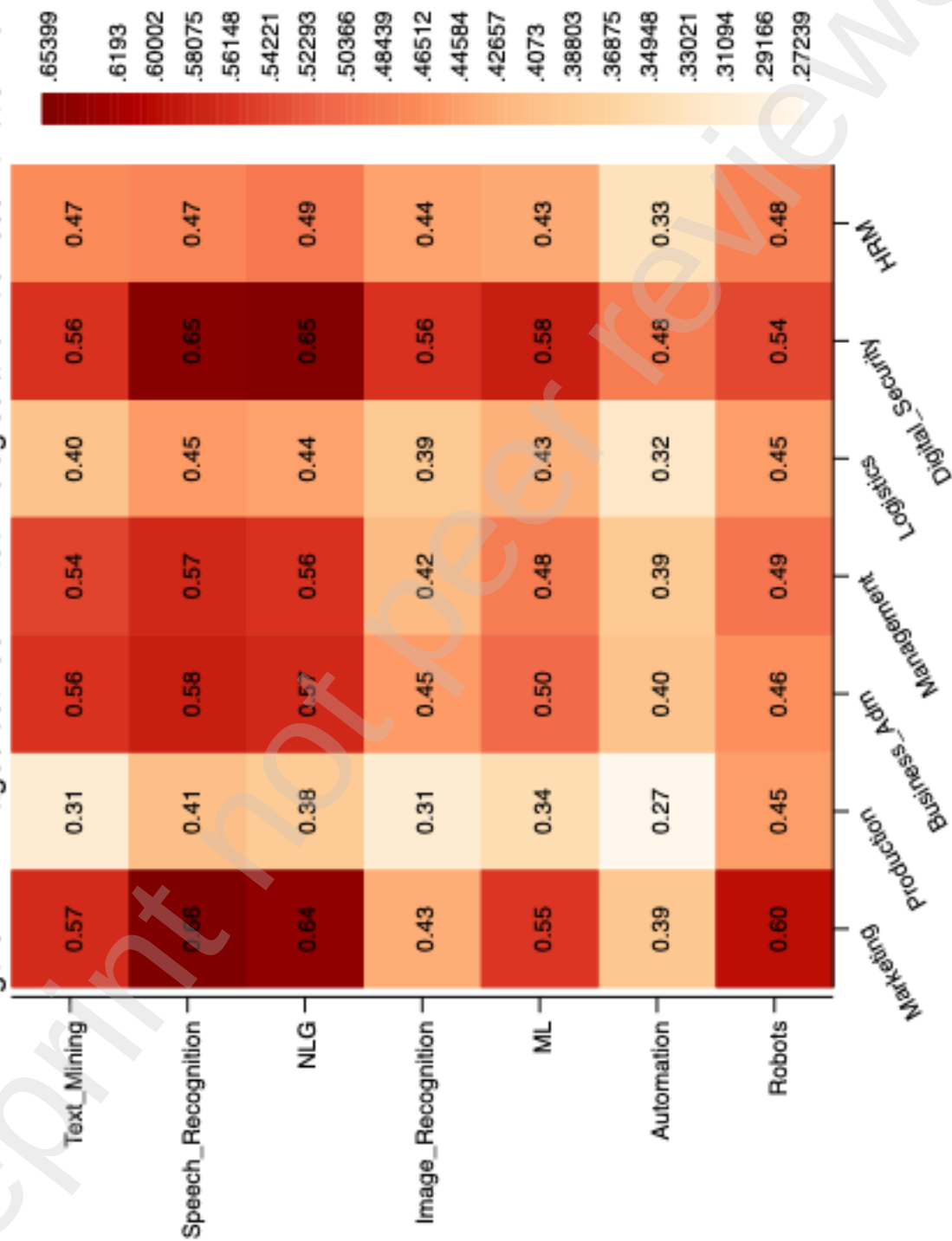


Figure 12. Linkages between AI technologies and Business Function 2023

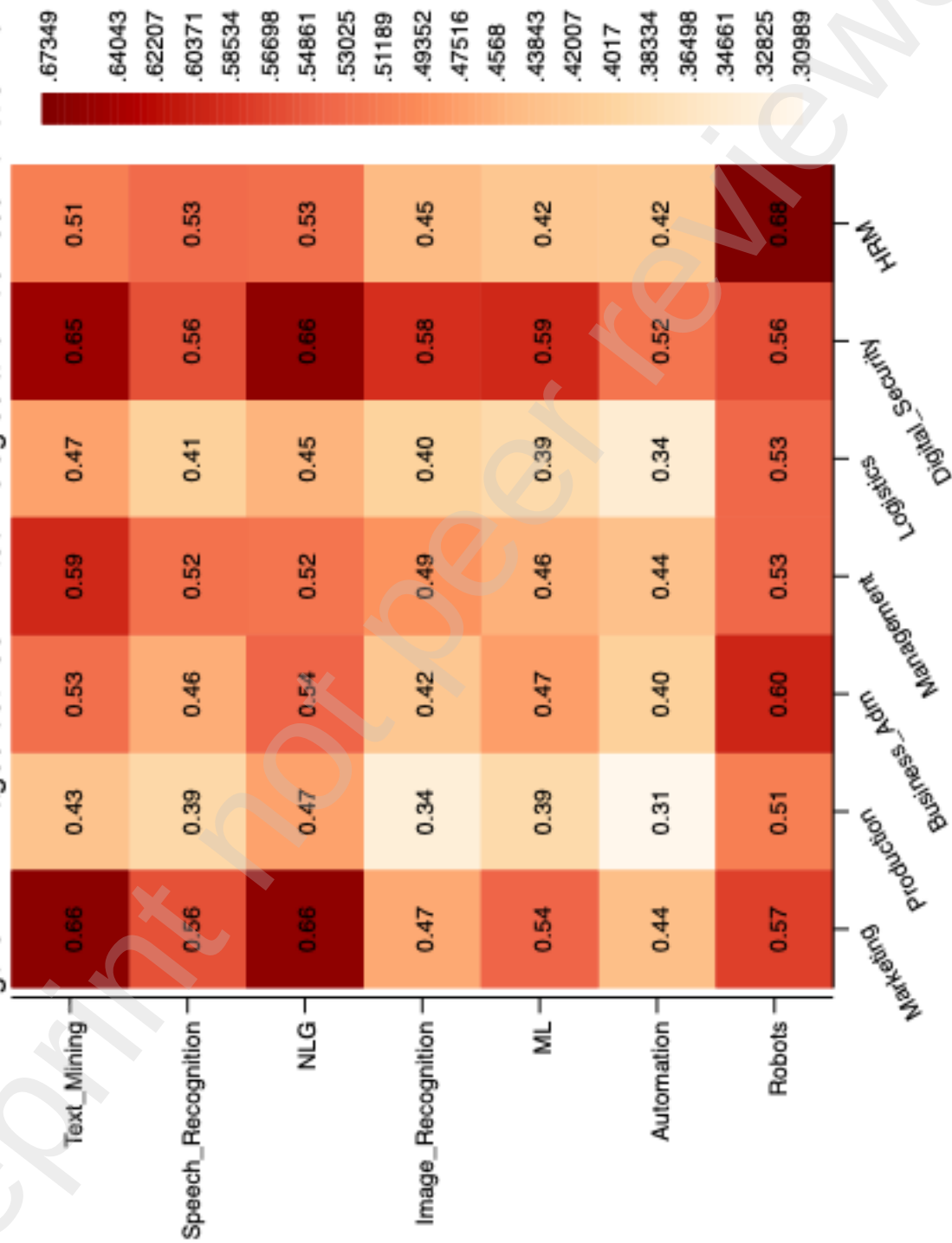


Figure 13. Linkages between AI technologies and Business Function 2024

